

Session 5: How AI Learns from Data

You know what AI is. You know **HOW** machines learn. Now we're going deep on the fuel that makes it all work - **DATA!**

Why does data matter so much? Why can good data make AI smart and bad data ruin it? Let's find out.



Objectives

By the end of this session, you will be able to:



Datasets

Explain what a dataset is



Train vs Test

Difference between training data and testing data



Overfitting

Spot when AI has memorized too much



Features

Tell what AI pays attention to - called **features**



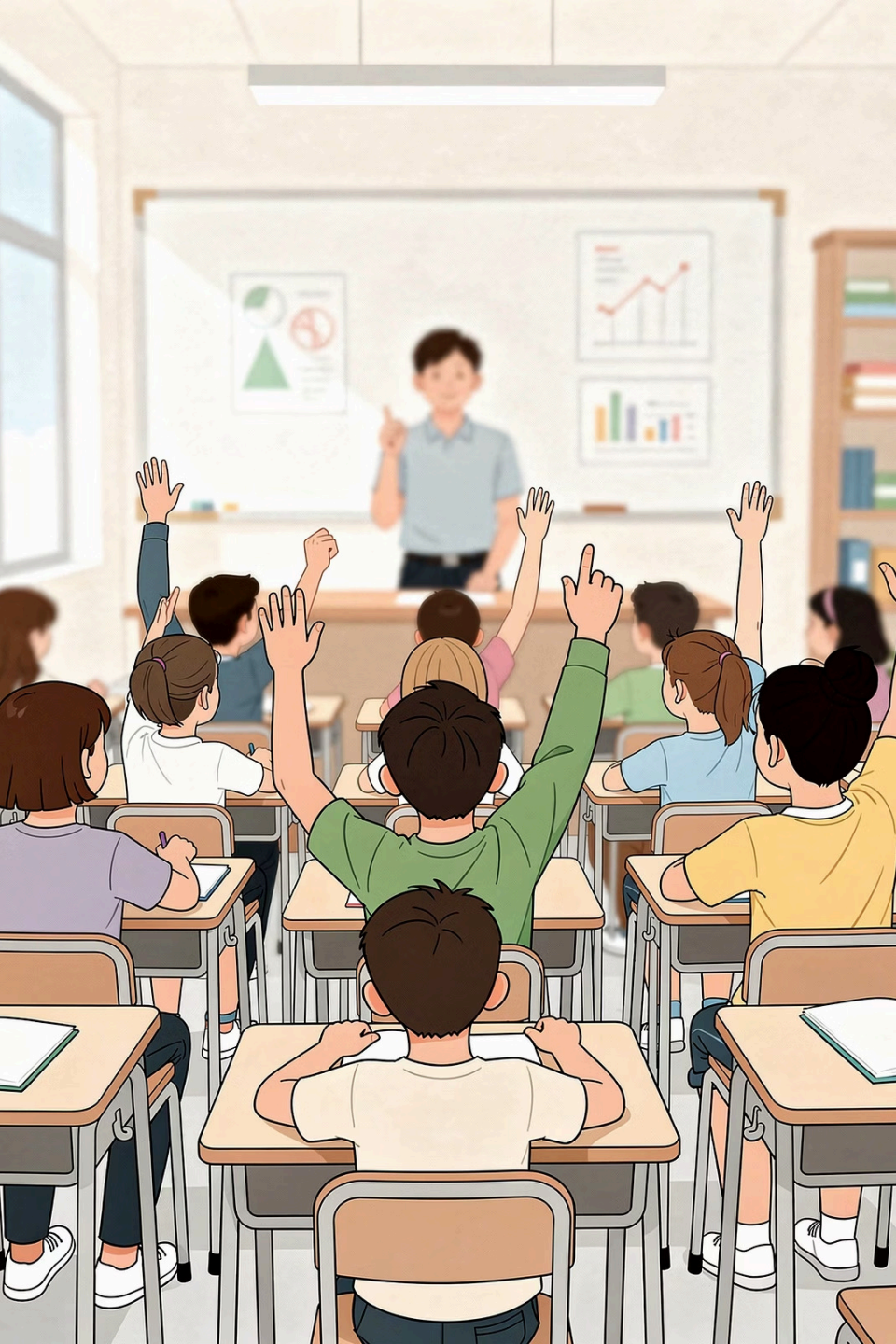
Patterns

Why AI must learn patterns, not just memorize



Bad Data

Recognize when data is bad or unfair



Recap & Homework

Welcome back! Let's start with your **ML Type Hunter**.

Share what you found:

Supervised Learning

Which app did you find? Learning **with answers** - labeled examples guide the AI.

Unsupervised Learning

Which app did you find? Learning **without answers** - just finding patterns!

Reinforcement Learning

Which app did you find? Learning with **rewards and trying again!**

The Big Question Today

Garbage In → Garbage Out

In this session we will find out exactly what that means - and how to spot it.

Identify

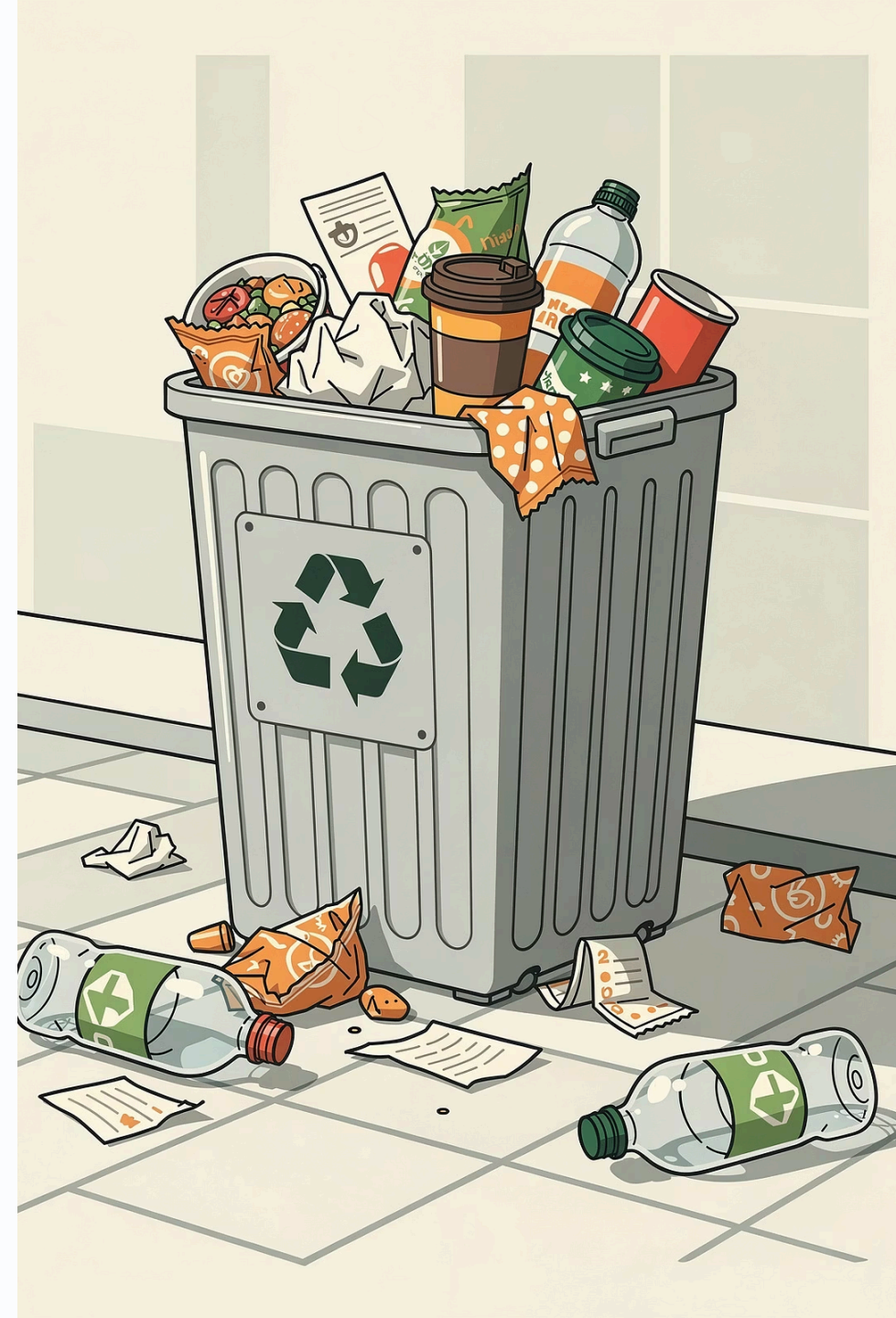
Learn what separates good data from bad data

Understand

Discover why data quality has real consequences

Spot It

Build the skills to recognize poor data in the wild





What is a Dataset?

A Dataset is just a **big collection of examples** for AI to learn from.

Think of it like:

Practice Flashcards

A box of labeled examples the AI studies over and over

Textbook of Examples

A structured collection of knowledge the AI reads through

Labeled Photo Folder

Images tagged with what they show so the AI learns to recognize them

Datasets in the Real World

Datasets come in many shapes and sizes - from spreadsheets of numbers to millions of images or sentences. Here are some everyday examples:

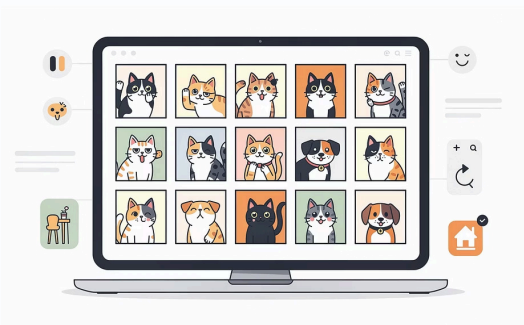


Image Datasets

Thousands of labeled photos teach AI to recognize objects, faces, and scenes.



Text Datasets

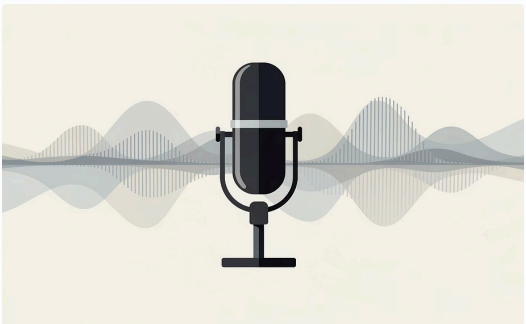
Books, articles, and conversations help language models understand and generate human text.

An illustration of a spreadsheet with a grid of numbers. The columns are labeled A through M, and the rows are numbered 0 through 12. The numbers are small and black, typical of a spreadsheet application.

	A	B	C	D	E	F	G	H	I	J	K	L	M
0	100	580	580	340	450	450	580	450	400	500	534	400	550
1	1800	1000	1500	1000	1560	1600	1700	1800	1500	1600	1800	1560	1800
2	200	250	160	370	160	100	160	170	120	160	1600	600	702
4	300	300	240	460	400	400	470	320	340	340	350	450	350
5	455	400	450	270	300	300	250	300	400	300	400	496	400
6	250	370	380	340	371	300	260	340	340	300	450	150	
6	200	300	300	200	300	300	200	300	300	300	240	340	300
7	160	361	160	220	250	320	340	361	200	300	260	160	151
8	260	370	100	340	371	370	370	370	380	360	320	380	
9	200	440	120	140	340	440	320	370	340	300	300	170	370
10	150	400	450	470	475	470	460	541	450	460	454	271	171
11	420	400	140	420	180	460	340	380	440	380	460	150	150
12	120	180	141	110	170	110	140	151	140	200	121	140	250
11	120	180	171	130	221	160	730	380	250	340	320	360	340
12	160	150	170	160	150	150	250	160	150	360	330	350	160
11	120	130	160	120	120	380	450	120	360	140	170	160	170
12	170	374	140	120	360	350	480	340	350	300	340	350	350

Tabular Datasets

Rows and columns of structured data power predictions in finance, healthcare, and more.



Audio Datasets

Recordings of speech, music, or environmental sounds used for voice assistants and audio classification.

What Are Features?

Features = the clues AI pays attention to.

Think of AI like a detective - it looks at clues to figure things out.

Example: How can AI tell apples from oranges? 🍏🍊

Clue (Feature)	🍏 Apple	🍊 Orange
Color	Red, green, yellow	Orange
Skin	Smooth	Bumpy
Shape	Round	Round
Has a stem?	Usually yes	Sometimes

📌 The AI learns: "Bumpy + orange color = probably orange. Smooth + red = probably apple."

Good Clues vs Bad Clues

Not all clues are useful! Some help. Some are just **noise**.

Helpful Clues - Spam Detection

- Words in the email ("FREE!" "Winner!")
- Who sent it
- How many links it has

Useless Clues - Spam Detection

- The color of the email background
- The font size

Picking the right clues is super important. **Bad clues = bad AI.**



Mini Activity 1 - Be a Clue Picker! 🐱🐶

You're teaching AI to tell **cats and dogs** apart. What clues would you use? For each clue, decide: **Useful** ✅ or **Useless** ❌

Clue	Useful or Useless?
Shape of the ears	?
Sound it makes	?
Color of the floor in the photo	?
Size and shape of the tail	?
Whether the photo was taken on a Tuesday	?
Whiskers	?

📌 Think carefully - some clues seem useful but are actually just noise!

Mini Activity 1 - Answers

Clue	Answer	Why
Shape of ears	✓ Useful	Cats: pointy / Dogs: floppy
Sound	✓ Useful	Meow vs bark
Floor color	✗ Useless	Has nothing to do with the animal!
Tail shape	✓ Useful	Cats: upright / Dogs: wagging
Tuesday photo	✗ Useless	Silly clue!
Whiskers	✓ Useful	Cats have very distinct whiskers - dogs much less so

👉 **Big idea:** AI is only as smart as the clues we give it. If we give bad clues, it learns bad patterns.



Training Data vs Testing Data

When teaching AI, we **DON'T** use all the data at once.

We split it into **2 groups**:



Training Data (~80%)

The examples AI **studies** to learn

Like your *practice problems* before a math test



Testing Data (~20%)

The examples AI **hasn't seen yet**

Like the *actual test* - new problems, no peeking!

Why the Split?

Why can't we just use **ALL** the data for training?

Because then we'd never know if the AI really learned - or just memorized.

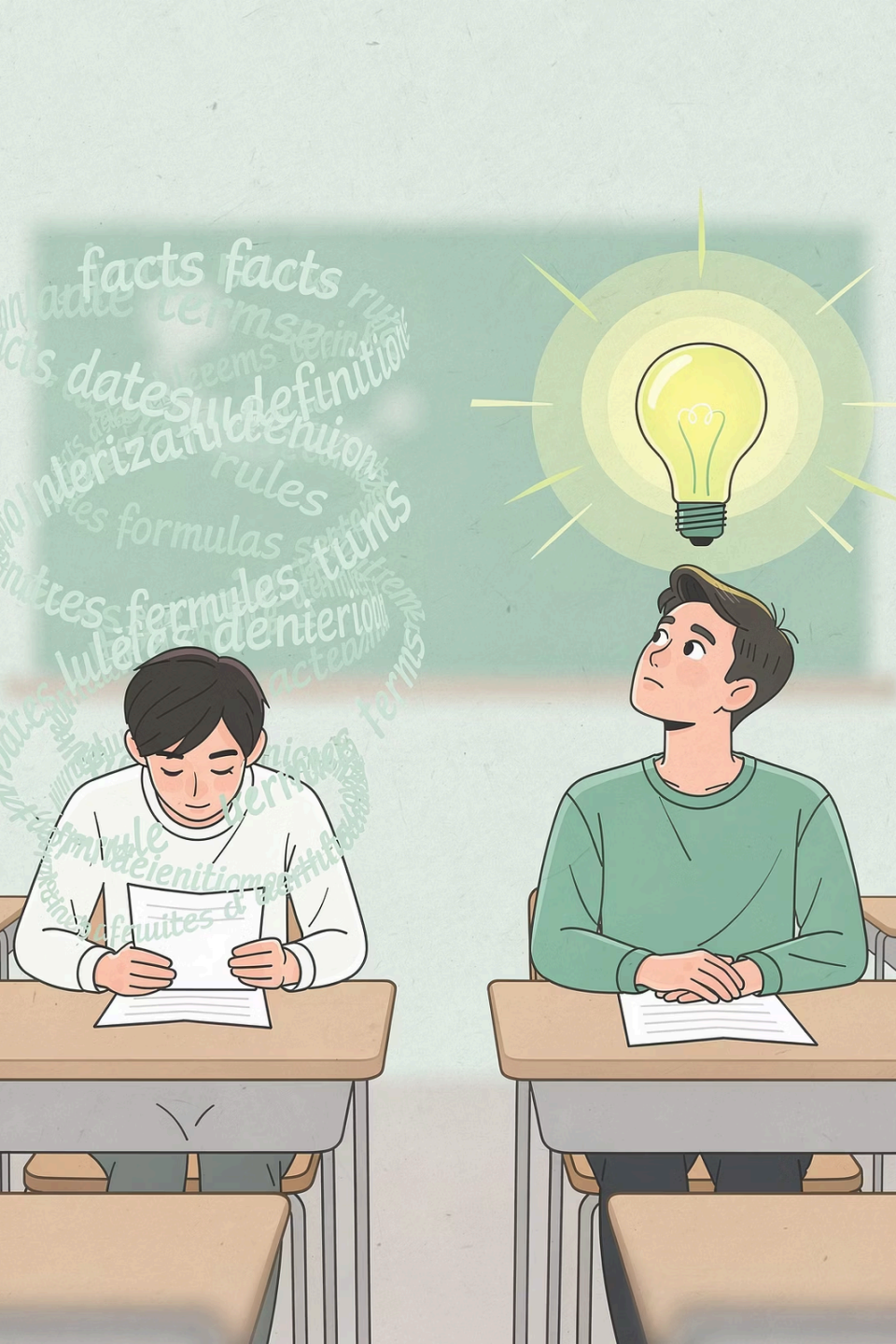
The Problem

You give the AI 100 photos to study. Then you test it on the **same** 100 photos. It gets them all right! Did it actually learn, or just memorize the answers?

The Real Test

Show it photos it has **NEVER seen before**. THAT tells you if it really learned. Engineers call this **splitting your dataset**. It's how every real AI project works.





Memorizing vs Learning

Here's one of the most important ideas in all of AI:



Memorizing

- Remembering exact answers
- Only works on stuff you've seen
- Like memorizing every word in a book
- Breaks when something changes



Learning

- Understanding the pattern
- Works on NEW stuff too
- Like understanding what the words mean
- Adapts to new situations



We want AI to **LEARN** - not just memorize.

The Goal - Generalization

Generalization = a fancy word for being able to handle **NEW stuff**.

You generalize all the time. Examples:



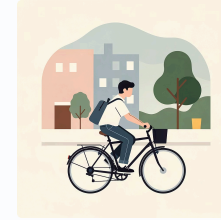
New Dog, Same Concept

You've never seen a Pomeranian before, but you can still tell it's a dog. You learned what a **dog is** - not just specific dogs.



New Pie, Same Category

You've never seen a cherry pie before, but you know it's pie. You understand the **concept of pie**, not just the ones you've tasted.



New Bike, Same Skill

You've never ridden *this specific* bike, but you can still ride it. You learned **how to ride**, not just one particular bicycle.

- ① That's generalization. You learned what a dog *is*, not just specific dogs you've seen. This is exactly what we want AI to do.

Overfitting

Overfitting = when AI memorizes the practice problems instead of learning the pattern.

Good AI (Learned)

- Sees: cats on couches, cats on rugs, cats outside
- Learns: "Cats have pointy ears, whiskers, tails"
- Can recognize a cat **ANYWHERE**

Overfit AI (Memorized)

- Sees: same cats, always on a **red couch**
- "Learns": "Cats live on red couches"
- Show it a cat on grass → "Nope, that's not a cat!"



Mini Activity 2 - Learning or Overfitting? 🤔

For each one, is the AI **LEARNING WELL** or **OVERFITTING**?

Scenario	Your Answer
AI trained on Golden Retrievers correctly identifies a Poodle as a dog	?
AI trained on sunny day driving crashes when it rains	?
AI that recognizes your face even with new glasses	?
AI that only recognizes cats on a specific couch	?
AI that can read handwriting it's never seen	?

📝 Think about whether the AI is responding to the **real pattern** or just surface details it memorized.

Mini Activity 2 - Answers

Scenario	Answer	Why
Recognizes a Poodle as a dog	✓ Learning	Got the IDEA of "dog"
Crashes in rain	⚠ Overfit	Only learned sunny days
Recognizes you with glasses	✓ Learning	Learned your face, not specific features
Only knows the couch cat	⚠ Overfit	Memorized the background
Reads new handwriting	✓ Learning	Got the pattern of letters

⚠ **Rule of thumb:** If the AI fails when something small changes, it probably overfit.



Problem 1 - Not Enough Data

What happens when AI doesn't have enough examples?

①

1 Photo of a Cat

AI thinks: "**ALL cats look exactly like THIS one.**" Anything different = "not a cat."

⑩

10 Photos

AI gets a little better, but still gets confused easily.



1,000+ Photos

AI sees **REAL patterns**. Way smarter. The more variety, the better.



The fewer examples, the more likely it overfits.

Problem 2 - Biased Data

Bias = when the data doesn't represent everyone fairly.

Example: A face recognition AI trained mostly on photos of light-skinned faces.

- Works great for some people
The AI performs well on faces similar to those it was trained on.
- Fails for people with darker skin
The AI never saw enough variety. Its data was unfair.

⊗ This isn't a small problem. It happens in real AI, to real people, every day. We'll go way deeper into this when we get to "**AI Bias & Fairness**" later in the course.





A Real Story - Self-Driving Cars in Rain

True story: A self-driving car was trained only on videos of sunny day driving. The day it rained for the first time... the car got confused.

What It Knew

Sunny roads, clear skies, dry pavement - that's all it had ever seen.

What Happened

It had never seen rain before. The training data lacked variety. No rain. No snow. No fog.

The Fix

Train it on every kind of weather, every kind of road, every kind of situation.

What Good Data Needs

For AI to be **smart and fair**, the data needs to be:



Enough

Lots of examples, not just a few



Clean

Labeled correctly, no mistakes - an "apple" photo isn't accidentally labeled "orange"



Varied

All kinds of examples - different lighting, angles, people, etc.



Fair

Represents everyone, not just one group

Miss any of these, and the AI gets bad - or unfair.

Hands-On Activity - Plan a Healthy/Unhealthy Food AI

Time to think like an AI builder. Your job: Plan an AI that can look at food and decide:

 **Healthy** or  **Unhealthy?**

01

What examples (data) would you give it?

Think about the variety of foods you'd include from different cultures and cuisines.

02

What clues (features) should it look at?

Sugar amount? Calories? Ingredients? Color? Shape?

03

How would you make sure your data is FAIR?

Does it represent foods from all cultures, not just one?



Hands-On - Step by Step

1 What's "healthy" and "unhealthy"?

Healthy: Apple, salad, grilled chicken, broccoli... **Unhealthy:** Soda, candy bar, chips, ice cream...

2 What clues (features) should the AI look at?

Sugar amount? Calories? Ingredients? Color? Shape?

3 How will you make sure the data is FAIR?

- 🍌 Did I include foods from cultures other than mine? (Mexican, Indian, Chinese, Japanese, Middle Eastern, African, etc.)
- 🥗 Did I include healthy AND unhealthy options from EACH of those cultures?
- 🏠 Did I include foods that LOOK different from what I usually eat?

4 How could it go wrong?

- Only photos of Western food → won't recognize sushi or curry
- Only fresh fruit → confused by dried fruit
- All on white plates → confused by other plates

⚠️ If you skipped any fairness checks - your AI will be biased.

What We Learned Today

Dataset

The box of examples AI learns from

Features

The clues AI pays attention to

Training vs Testing Data

Practice vs the real test

Memorizing vs Learning

AI must understand patterns, not just remember

Generalization

The goal - works on NEW stuff

Overfitting

When AI memorizes too much

Bias

When data isn't fair to everyone

✓ **Big idea: Good data = Good AI. Bad data = Bad AI. Always.**



Your Homework - Prepare for Cats vs Dogs!

Next session, we'll train a **REAL AI** to tell cats from dogs.

Tonight, plan it out:

Task 1: What clues (features) would you use?

Write down **at least 4 clues**. Examples:

- Ears (pointy or floppy?)
- Face shape
- Tail
- Sound
- Size

Task 2: What might trick the AI?

Write down **at least 2 things** that could confuse it. Examples:

- Photos taken in the dark
- Photos where the animal is hiding behind something
- A cat lying down like a dog

Next Time - Session 6: Training Your Own AI Model

You've been planning. Next session, you **BUILD**.



Explore Kaggle

Where real AI engineers find datasets



Train a Classifier

Build a Cats vs Dogs classifier together



Run Experiments

What happens with little data vs lots?



Watch Overfitting

See overfitting happen in real time

